

WATERWALKER · CALCULATION NOTE

WaterWalker sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (geometry, mass, push force on firm ground, slopes and sand, assist, brakes, frame, axles, turning, cost, requirement table)

WaterWalker sizing calculations

On paper, the first prototype (no assist, with mounting points) carries 80 L, turns in 3.7 m, lifts containers only 364 mm and costs about \$430. But it weighs about 40.8 kg empty, not the 31 kg estimated at TRL 2, and that extra mass pushes R2 (firm ground) and R4 (10 % climb) just past their targets. **Four requirements are not met:** R2 (61.4 N against 60 N), R3 (loose sand, 242 to 368 N against 150 N), R4 (183 N against 180 N) and R9 (40.8 kg against 35 kg). Six are at risk: R1, R6, R7, R10, R11 and R12. The calculations also found two layout faults in the TRL 2 concept that are now fixed in the model: the cantilevered rear stub axles were overstressed, and the swivelling front tires would hit the front risers.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the prices from `bom/bom.csv`, so the note, the model, drawing WWK-DWG-001 and the BOM share one source. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Load	Four 20 L jerrycans, 80 kg of water plus 4 x 1.1 kg; or two 20 L clay pots of 8 kg each	WWK-REQ-001 R1
Wheels	26 in, ETRTO 559 rim, 2.1 in (54 mm) tires, radius 333.5 mm; 2.4 in (61 mm) also fits, radius 340.5 mm	Decided, WWK-DDR-001 item 1
Hubs	Front-type 100 mm over-locknut hubs on all four wheels; 70 mm drums on the rear pair	Proposed, WWK-DDR-001 item 9
Rolling resistance, firm dirt path	Crr 0.02 to 0.05	Typical for wide bicycle tires on packed to rough dirt
Rolling resistance, loose sand	Sinkage 25 to 55 mm, giving Crr 0.197 to 0.300 (section 4)	To be measured in the field
Grade	10 % climb and descent (R4); 20 % parking (R10)	WWK-REQ-001
Steel	Mild steel, yield 235 MPa, 7,850 kg/m ³ , G = 79 GPa	S235 class; ERW tube is often stronger
Dynamic factor	2.5 g on static loads	Ruts, stones and kerbs at walking speed
Target factor on yield	1.5 or more at 2.5 g	Paper design margin
Component masses	Rim 0.55 kg, spokes 0.30, puncture-resistant tire 0.85, thorn-resistant tube 0.45, liner 0.20, drum hub 0.75, plain hub 0.35, rigid fork 1.10 each	Typical catalogue values for utility parts
Drum brake	70 mm drum, brake factor C* = 0.8, cam ratio 5, lever ratio 4, cable efficiency 0.8, one lever to both drums	Typical cable-operated drums
Assist motor (later)	250 W, 48 V geared hub motor, 40 N·m at low speed; 40 % efficient at 17 to 29 rpm	Assumed; geared hub motors are rated at much higher speeds
SwapCell pack	46.8 V nominal, 452 Wh at minimum cell capacity, 2.85 kg, 20 A continuous, 15 A in legacy mode	SwapCell interface v0.3 and SWC-CAL-001
Household use	100 L per day, 3 km round trip at 4 km/h	WWK-PRB-001

■ 2. Geometry and layout

The layout is set by three limits that squeeze each other: overall width (R6, 900 mm), clear walking width (R7, 600 mm) and 100 mm bicycle hubs held on both sides. With a 770 mm track the carrier is 898 mm wide over the axle nuts and 610 mm clear between the side rails, so both R6 and R7 hold, each by less than 2 %. The overall length is 2,197 mm (2,211 mm with 2.4 in tires).

Two TRL 2 layout faults were found and fixed in the model:

- **Rear axles.** The TRL 2 concept hung each rear wheel on a stub axle outboard of the side rail. A standard M10 bicycle axle cantilevered 40 mm carries about 32.7 N·m at 2.5 g, which is about 472 MPa, beyond what a bicycle

axle survives in fatigue. Holding each rear wheel in a fixed rigid bicycle fork with 100 mm dropouts cuts this to about 89 MPa and uses the same fork as the front casters (proposed, WWK-DDR-001 item 9).

- **Caster sweep.** A swivelling front wheel sweeps a circle of 400.5 mm radius about its swivel axis (60 mm trail plus the 2.4 in tire radius). The TRL 2 risers sat inside that circle, so the tire would hit them after only a few degrees of steer. The risers now stand at $x = 1,150$ mm and the wheelbase is 1,530 mm, which leaves 19.5 mm of clearance (proposed, WWK-DDR-001 item 10).

The cradle is 780 x 430 mm outside and 768 mm long inside. Two jerrycans in line (720 mm) leave 44 mm; two 380 mm clay pots leave only 8 mm, so the pot case (R1) is at risk until real pot sizes are known. The heaviest lift is over the side rail at 364 mm (R8).

■ 3. Mass

Table 2. Mass roll-up, first prototype.

Part	Mass (kg)	Basis
Frame	14.62	8.25 m of tube from <code>frame_members()</code> in the model: 5.69 m of 40 x 30 x 1.5 mm (8.98 kg), 0.97 m of 30 x 30 x 1.5 mm (1.31 kg), 0.89 m of 25 x 25 x 1.5 mm (0.99 kg), 0.70 m of 20 x 20 x 1.5 mm (0.61 kg); head tubes, tabs, receiver plate, welds and paint
Rear wheels with drum hubs (2)	6.20	Table 1
Front wheels (2)	5.40	Table 1
Forks (4)	4.40	Table 1
Headsets, clamps and swivel locks	0.60	Estimate
Hip bar, posts, pad and grips	2.20	Estimate
Cradle	5.08	Plywood floor 1.81, rubber pad 1.48, walls 1.39, straps 0.40
Brake lever, latch and cables	0.45	Estimate
Skirt guards (2)	1.29	3 mm HDPE
Hardware and reflectors	0.60	Estimate
Total, empty	40.84 (90.0 lb)	R9 limit 35 kg

The assist kit adds 6.50 kg (motor wheel less the drum hub, disc brake, controller, sensor, class V1 receiver and the 2.85 kg SwapCell pack), for 47.34 kg against the 42 kg limit. Loaded with four jerrycans, the carrier weighs 125.2 kg, or 131.7 kg with the assist kit. The loaded centre of mass is 716 mm ahead of the rear axle and 364 mm above the ground, so the front wheels carry 46.8 % of the load: about 287 N per front wheel and 327 N per rear wheel.

R9 is **not met**, by 5.8 kg. The TRL 2 figure of 31 kg left out the tires, tubes and liners (1.5 kg per wheel), the fourth pair of forks, and the heavier frame section that the strength check calls for. Four mass options (not applied to the model) save 5.21 kg: a 1.2 mm wall on the main tube (1.80 kg), a thinner cradle (1.81 kg), tires without a puncture belt (0.80 kg) and no tire liners (0.80 kg). Even with all four the carrier weighs 35.63 kg, and the last two trade weight for punctures on thorny paths (WWK-DDR-001 item 12).

■ 4. Push force

Push force is $F = m g (C_{rr} \cos \alpha + \sin \alpha)$ for a grade angle α .

On a firm, level dirt path the loaded carrier needs 24.6 to 61.4 N, which is 25 to 61 W of human power at 1.0 m/s. R2 is **not met** at the rough end of the range, by 1.4 N. Climbing a 10 % grade on firm ground takes 146.7 to 183.4 N; R4 is **not met** at the rough end, by 3.4 N. Both follow from the mass: with all four mass options the figures fall to 58.9 N and 175.8 N, which would be at risk rather than not met.

Loose sand. A rigid wheel that sinks z into soil meets it at an entry angle θ , where $\cos \theta = 1 - 2z/D$, and the soil reaction passes through the axle, so the rolling resistance coefficient is $\tan(\theta/2)$. For a 667 mm wheel, a sinkage of 25 to 55 mm gives C_{rr} 0.197 to 0.300, which matches the 0.2 to 0.3 used at TRL 2. The loaded carrier then needs **242 to 368 N**, which is 1.6 to 2.5 times the 150 N target. R3 is **not met**, and the first prototype has no assist.

Table 3. Effect of tire size on the sand push force (loaded, unassisted).

Tire	C_{rr} factor	Sand push force (N)
26 x 2.1 in	1.00	242 to 368
26 x 2.4 in	0.95	230 to 350
26 x 3.0 in	0.86	209 to 317
26 x 4.0 in (fat bike)	0.75	182 to 276
28 x 2.0 in	0.97	236 to 358

The factors scale sinkage with Bekker's pressure-sinkage relation for a soil whose frictional modulus dominates, with a sinkage exponent n of 1.1, typical of dry sand, so sinkage varies as $(W / (b \sqrt{D}))^{2/(2n+1)}$, and C_{rr} as $\sqrt{z/D}$. They are relative only. Even fat-bike tires leave the push force well above 150 N, so the TRL 2 question of whether wider tires could avoid assist is answered: they cannot. Carrying only two jerrycans on a sandy stretch (83.0 kg) still takes 161 to 244 N.

On a 10 % descent the carrier pulls downhill with 61 to 98 N more than rolling resistance absorbs.

■ 5. Assist kit (later prototype)

The assist is not in the first prototype (WWK-DDR-001 item 2); this section checks that the mounting points suit a later kit built to SwapCell interface v0.3.

One 250 W geared hub motor at 40 N·m gives 120 N of thrust at the 333.5 mm wheel radius. On loose sand the user would still push 135 to 267 N, so one motor does not meet R3. Meeting R3 on the loosest sand with the assisted mass takes 237 N of thrust, 79 N·m at the wheels: about 40 N·m from each of two motors, one in each rear wheel.

With two motors the user pushes 15 to 148 N. Two motors also cancel the yaw from one-sided drive: one motor makes a 46 N·m yaw moment, which locked casters resist with 30 N of side force (5 % of the front load) but free casters leave to the user as a 72 N difference between the hands.

At walking pace the wheel turns at only 17.2 rpm (0.6 m/s) to 28.6 rpm (1.0 m/s), far below where hub motors are efficient. At an assumed 40 % efficiency, one motor on sand uses 180 W and turns 108 W into heat; two motors at the R3 thrust use 356 W (7.6 A from the 46.8 V pack) and make 214 W of heat, so motor temperature is at risk and needs a thermal check before any assist build. Energy use is about 83 Wh/km with one motor and 165 Wh/km with two on loose sand, so a SwapCell pack (452 Wh) gives about 5.4 km or 2.7 km of assisted sand. The current is well inside the pack's 15 A legacy-mode limit, so a simple controller without CAN can use the pack through the INTERLOCK coding resistor alone (interface v0.3 item W).

■ 6. Brakes and parking (R10)

Parking the assisted carrier (131.7 kg) on a 20 % grade needs 253 N at the tires, or 42.3 N·m in each rear drum. With the assumed drum and cable ratios that needs about 189 N at the lever when the latch is set (179 N unassisted), which is a hard squeeze for many users. Traction is the second limit: parked facing downhill, the braked rear wheels carry only 614 N, so the tires need a friction coefficient of 0.41 (0.35 facing uphill). That holds on firm dirt but not reliably on loose gravel or sand. R10 is therefore **at risk**. A positive lock pin through a rear wheel removes the hand-force problem but not the traction limit (WWK-DDR-001 item 13).

Holding speed on a 10 % descent takes about 73 N at the lever, and stopping from 1.0 m/s within 1.0 m (0.5 m/s²) with the assisted mass takes about 126 N. The service brake meets R4's descent clause on paper, but sustained 73 N on long descents is tiring.

■ 7. Frame, axles and hip bar

One side rail is treated as a beam on the rear axle and the front tire contact, loaded through two cradle hangers (219 N each, static) and its share of frame mass. Table 4 gives the peak bending stresses at 2.5 g for the 40 x 30 x 1.5 mm rectangular section ($Z = 2,302 \text{ mm}^3$).

Table 4. Frame stresses at 2.5 g.

Location	Moment, static (N·m)	Moment at 2.5 g (N·m)	Stress (MPa)	Factor on yield
Side rail at the rear hanger	97	243	106	2.22
Side rail at the front riser	92	231	100	2.34
Caster arm root	103	258	112	2.09

The TRL 2 section (30 x 30 x 1.5 mm square) would reach 167 MPa at the caster arm root, a factor of only 1.41, so the main members are upsized (proposed, WWK-DDR-001 item 11). A 1.2 mm wall would give 137 MPa (factor 1.72). Weld fatigue is **at risk**: a $\pm 0.75 \text{ g}$ swing about 1 g gives a 63 MPa stress range at the rear hanger, close to the 71 MPa of a typical fillet-welded detail at 2 million cycles. It cannot be verified without testing.

With one front wheel unloaded in a rut, the frame carries 221 N·m of torsion. If one rail took it all the shear stress would be 67 MPa; shared by both rails the frame twists about 1.71 degrees, so a front wheel stays on the ground over a rut about 23 mm deep before the carrier rests on three wheels.

The hip bar uprights (30 x 30 x 1.5 mm) carry a 400 N push shared between them at 616 mm above the rails: 123 N·m and 80 MPa each, a factor of 2.9.

■ 8. Turning and stability

With the casters trailing tangentially and the 2.4 in tires, a turn about the inner rear wheel sweeps a circle of 3.66 m, and a spin about the rear axle centre (one rear wheel rolling backward) sweeps 3.41 m. R5 is met.

The static side tip angle, loaded, is 47 degrees (half track 385 mm, centre of mass 364 mm). The static tip angles over the front and rear axles are 66 and 63 degrees. These are static angles; a wheel dropping into a rut on a cross-slope still needs a dynamic check in the field.

■ 9. Water delivered and cost

A household using 100 L per day needs 5 head-carried trips or 2 WaterWalker trips (1.25 rounded up). On a 3 km round trip at 4 km/h that is 3.75 h against 1.50 h of walking, if walking speed is similar; sandy routes are slower.

The priced BOM gives **\$430.00** for the first prototype (no assist, with mounting points) against the \$450 budget, a margin of \$20.00, so R11 is at risk. The assist kit (hub motor, sensor and class V1 receiver) adds \$260.00; the SwapCell pack is priced once in the SwapCell BOM and excluded here.

■ 10. Results against every requirement

Status rule: **met** means the value is within 95 % of the limit; **at risk** means within the limit but with less than 5 % margin, or dependent on an unverified assumption; **not met** means past the limit.

Table 5. Requirement results (WWK-REQ-001 v0.3).

ID	Target	Value	Status
R1	80 L in four jerrycans or 40 L in two clay pots; payload 90 kg or less	84.4 kg with jerrycans, 56 kg with pots; spare length 44 mm (jerrycans), 8 mm (pots)	At risk (pot size)
R2	60 N or less, firm level path	24.6 to 61.4 N	Not met
R3	150 N or less, loose sand (with assist if fitted)	242 to 368 N unassisted; 135 to 267 N with one motor; 15 to 148 N with two	Not met (first prototype has no assist)
R4	180 N or less on a 10 % climb; controlled descent	146.7 to 183.4 N climb; 73 N at the lever to hold on the descent	Not met (climb)
R5	4.0 m turning circle or less	3.66 m about the inner rear wheel; 3.41 m spin	Met
R6	900 mm wide or less	898 mm over the axle nuts	At risk
R7	No step-over; walking width 600 mm or more; hip bar 850 to 1,050 mm; skirt guards	0 mm; 610 mm; 850 to 1,050 mm; guards on both rear wheels	At risk (walking width margin 10 mm)
R8	Lift 450 mm or less, from either side	364 mm	Met
R9	Empty 35 kg or less (42 kg with assist)	40.8 kg (47.3 kg)	Not met
R10	Service brake while walking; park rated gross mass on 20 %	Lever on the grip; 189 N at the latch; tire friction 0.41 needed	At risk
R11	First prototype (no assist, with mounting points) \$450 or less	\$430	At risk
R12	Wear parts are standard 26 in bicycle parts	26 in rims, tires, tubes, spokes, cables, headsets; 100 mm drum hubs are less common than rim-brake hubs in rural markets	At risk
R13	Assist-ready to SwapCell interface v0.3	Receiver plate, torque-arm tab and sensor boss in the model; 7.6 A peak against the 15 A legacy limit; latch class V1 retention needs a test	Not verifiable at TRL 3

Summary: 2 met, 6 at risk, 4 not met (R2, R3, R4, R9), 1 not verifiable at TRL 3.

■ 11. Corrections to TRL 2 figures

The TRL 2 figures in WWK-PRC-001 v0.2 were checked against this script and corrected in v0.3: empty mass 31 kg to 40.8 kg (47.3 kg with assist); loaded mass 115 kg to 125.2 kg; firm push force 25 to 55 N to 24.6 to 61.4 N; sand push force 225 to 340 N to 242 to 368 N; 10 % climb 135 to 170 N to 146.7 to 183.4 N; hold-back on the descent 55 to 90 N to 61 to 98 N; parking force 250 N to 253 N; turning circle 3.8 m to 3.66 m; width 860 mm to 898 mm; walking width 660 mm to 610 mm; lift height 350 mm to 364 mm; side tip angle 45 degrees to 47 degrees; parts cost about \$300 to \$430; assist energy 70 Wh/km to 83 Wh/km; and the assist pack is 46.8 V (SwapCell 13S), not 36 V.

Safety: These are paper calculations. The brake, parking and frame figures depend on assumed drum ratios, tire friction and a 2.5 g dynamic factor, none of which has been measured. Nothing in this note makes the carrier safe to build or load. TRL 4 testing is on hold by Amish's instruction.